NATURAL LANGUAGE PROCESSING

Comparative Study of NLP Models and Algorithms

Understanding approaches, strengths, limitations, and applications

Name: A. Lakshmipathi

RegNo: 192472323

Course: Natural Language Processing

Topic: NLP Models and Algorithms

RULE-BASED • MACHINE LEARNING • DEEP LEARNING

1. Introduction to NLP Models

Natural Language Processing (NLP) enables computers to process, understand, and generate human language.

Rule-Based Models

Use hand-written grammar rules, dictionaries, and patterns.

Best for: clear, limited tasks.

Machine Learning

Learn patterns from labelled examples using statistical features.

Best for: classification tasks.

Deep Learning

Neural networks learn complex language patterns from data.

Best for: context-rich tasks.

Common NLP tasks

- Sentiment analysis and spam detection
- Machine translation, summarization, and chatbots

2. Popular NLP Algorithms and Models

Naive Bayes

Probability-based; often used for spam filtering and text classification.

Support Vector Machine

Separates text categories using a decision boundary over feature vectors.

LSTM / RNN

Neural models that process sequences and capture order in text.

BERT

Transformer model that uses surrounding context to represent words.

GPT-Style Models

Transformer language models trained to predict and generate text.

3. Comparison: Strengths and Limitations

Approach	Strengths	Limitations
Rule-based	Easy to explain; uses expert rules	Hard to maintain; weak on unseen patterns
Traditional ML	Fast with suitable features	Needs feature design and labelled data
Deep learning	Learns complex patterns and context	Needs more data and computing resources
Transformers	Strong contextual understanding; versatile	Can be costly; may produce incorrect outputs

No single model is best for every task: performance depends on data, goal, cost, and language.

4. Applications and Conclusion

Model selection should match the problem, available data, and computing resources.

Applications

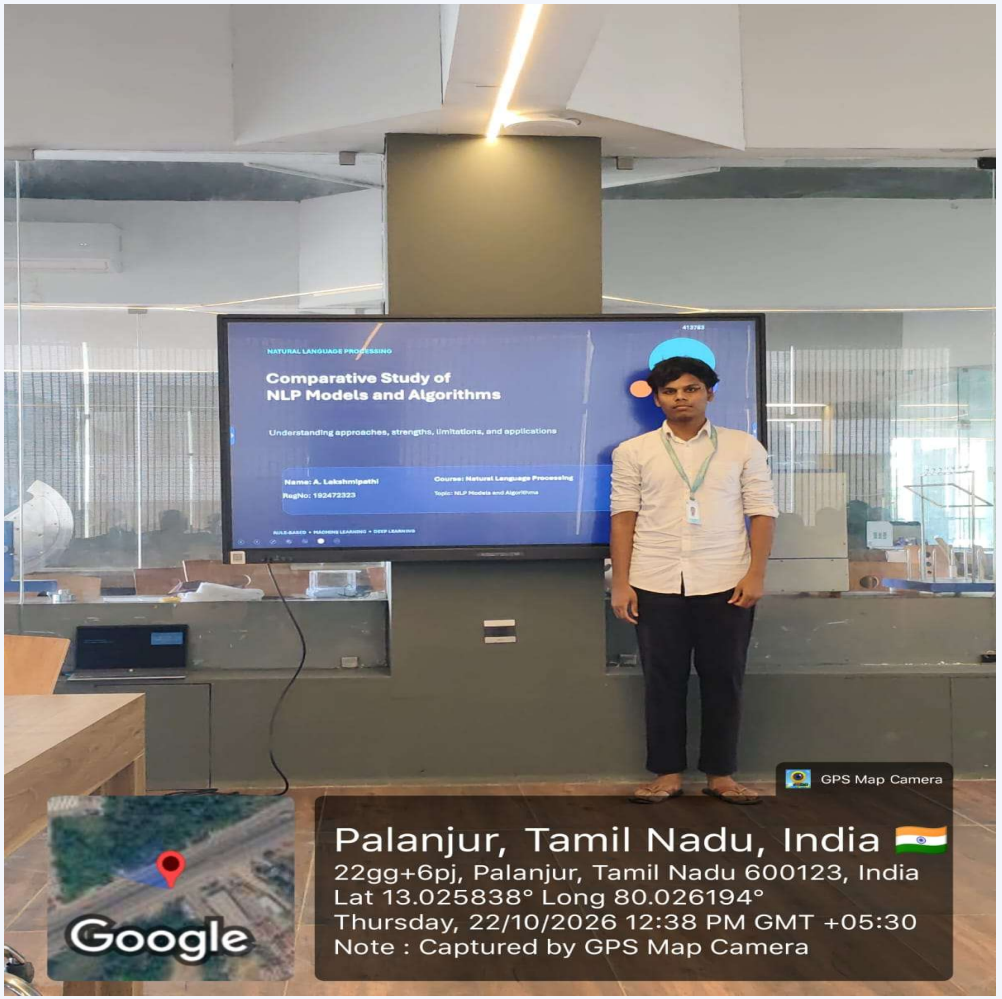
- Spam and sentiment classification
- Translation and text summarization
- Question answering and chatbots
- Information extraction and search

Conclusion

- Rule-based systems are simple and interpretable.
- Traditional ML can be efficient for focused tasks.
- Deep learning and transformers capture richer patterns.
- Test models for accuracy, fairness, and reliability.

Key takeaway: Choose the simplest model that meets the task's quality requirements.

Thank You


NATURAL LANGUAGE PROCESSING
Comparative Study of
NLP Models and Algorithms
Understanding approaches, strengths, limitations, and applications
Name: A. Lakshmipathi
RegNo: 192472323
Course: Natural Language Processing
Topic: NLP Models and Algorithms
RULE-BASED • MACHINE LEARNING • DEEP LEARNING
GPS Map Camera
Palanjur, Tamil Nadu, India
22gg+6pj, Palanjur, Tamil Nadu 600123, India
Lat 13.025838° Long 80.026194°
Thursday, 22/10/2026 12:38 PM GMT +05:30
Note : Captured by GPS Map Camera
Google